INCH-POUND

MIL-P-52434C(ME)
30 January 1992
SUPERSEDING
MIL-P-52434B(ME)
11 January 1979

MILITARY SPECIFICATION

PRINTING PRESS, OFFSET, LITHOGRAPHIC

(MOTOR DRIVEN, 24- BY 32-1/4-INCH MAXIMUM SHEET SIZE)

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a motor-driven, single-color, lithographic, offset printing press.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3610

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SPECIFICATIONS

FEDERAL

GG-P-455	- Plates and Foils, Photographic (Photosensitive Anodized Aluminum).
UU-T-81	- Tags, Shipping and Stock.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape, Packaging, Waterproof.

MILITARY

MIL-V-3	- Valves, Fittings, and Flanges (Except for Systems Indicated Herein): Packaging of.
MIL-C-104	- Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-V-173	- Varnish, Moisture and Fungus Resistant (for Treatment of Communications, Electronics and Associated Equipment).
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-H-775	- Hose, Hose Assemblies; Rubber, Plastics, Fabric, or Metal (Including Tubing); and Fittings, Nozzles and Strainers, Packaging of.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-C-3600	- Compressor, Rotary, Power-Driven; and Compressor, Reciprocating, Power-Driven: Air and Gas (Except Oxygen and Refrigerant), Packaging of.
MIL-I-3606	- Ink, Printing: Lithographic Offset, with Drier.
MIL-C-3774	- Crates, Wood; Open 12,000- and 16,000-Pound Capacity.
MIL-G-3859	- Grease Guns, Hand Operated, Lever, Push and Screw Type.
MIL-C-5501	- Caps and Plugs, and Plug, Protective, Dust and Moisture Seal.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power-Driven, for Water, Packaging of.
MIL-V-13811	- Varnish, Waterproofing, Electrical, Ignition.
MIL-E-16298	- Electric Machines having Rotating Parts and Associated Repair Parts: Packaging of.
MIL-G-25013	- Grease, Aircraft, Ball and Roller Bearing, NATO Code Number G-372, Metric.

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MIL-P-43027	- Paper, Chart, Map and Book Lithographic Finish, Coated and Uncoated.
MIL-C-52950	- Crates, Wood, Open and Covered.
MIL-P-53030	- Primer Coating, Epoxy, Water Reducible, Lead and Chromate Free.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Military Materiel (Metric).
MIL-STD-1791	- Design for Internal Aerial Delivery in Fixed Wing Aircraft.
MS15003	- Fittings, Lubrication (Hydraulic) Surface Check, 1/8 Pipe Threads, Steel, Type III.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Section 6 - Rules Governing the Loading of Department of Defense Materiel on the Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 50 F Street N.W. Washington D.C. 20001)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG1 - Motors and Generators.

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(Application for copies should be addressed to the National Electrical Manufacturers Association, 2101 L Street, N.W., Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, One Batterymarch Park, P.O. Box 9101, Quincy, MA 02269-9101.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION INC. AGENT.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., ATTN: Traffic Order Section, 2200 Mill Rd, Alexandria, VA 22314.)

SOUTH COAST AIR QUALITY MANAGEMENT CONTROL DISTRICT (SCAQMCD)

Rule 442 - Usage of Solvents.

(Application for copies should be addressed to the South Coast Air Quality Management Control District, 9150 East Flair Drive, El Monte, CA 91731.)

UNIFORM CLASSIFICATION COMMITTEE AGENT

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The press, consisting of a base and frame supporting the feeder and suction head, and inking, dampening, delivery, and printing assemblies, shall be capable of reproducing line, halftone, and color subjects in hairline register by the offset printing process.

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3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.3 Material. The material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subjected to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The press shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the press may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the press may be newly fabricated from recovered materials to the maximum extent practicable, provided the press produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the press.

3.4 Manufacturer's product lines. The press and its components shall be of the latest model(s) of the contractor (and any subcontractors). Product improvements incorporated in the manufacturer's production units to meet military requirements are acceptable. The press shall be equipped with all components necessary to enable the unit to function reliably and efficiently in sustained operation. All components and optional items shall be represented in the various manufacturer's current technical data.

3.5 Design. The press shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent conditions hazardous to personnel or deleterious to equipment.

3.6 Maintainability. All assemblies and installed attachments shall be accessible for maintenance and repair without the removal of other assemblies and installed attachments. Covers or plates which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with quick-disconnect fastenings. Oil drip pan shall be provided.

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3.7 Transportability. The press shall be capable of being transported by military or commercial trailers, trains, marine vessels, aircraft and must be capable of withstanding the impact forces encountered in shipment, without damage or permanent deformation. The press shall be equipped with tiedown or slinging provisions.

3.7.1 Tiedown provisions. The tiedown provisions shall conform to MIL-STD-209, class 2 or 3, type IV equipment, and to MIL-STD-1791 for equipment restraint criteria. The tiedown provisions shall satisfactorily complete the pull testing as specified without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components.

3.7.2 Slinging provisions. The slinging provisions shall conform to MIL-STD-209, class 1 or 3, type IV. The provisions shall enable the complete press to be lifted in the normal operating position. The provisions shall be located so that not less than one inch clearance is maintained between slings and all exterior parts and shall be fastened to members which will withstand stresses in the amount and direction of pull specified for the provisions without weld failure, permanent deformation, cracking, loosening, or breaking of the provision or its connecting structural components. Slinging provisions may also be used as tiedown provisions when such provisions meet the requirements specified in 3.7.1. All slinging/tiedown provisions shall be labeled 'LIFT', 'TIEDOWN' or 'LIFT TIEDOWN', as appropriate, in 1-inch (2.54 cm) high letters.

3.7.3 Air transportability. The press shall meet the requirements of MIL-STD-1791 for air transport.

3.7.4 Rail transportability. The press shall be rail transportable in CONUS and NATO countries without restrictions. The press shall have dimensional profile within the Gabarit International de Chargement (GIC) in accordance with the AAR, section 6, outline diagram when loaded on a 50-inch (127 cm) high rail car. The press shall be capable of withstanding shock loads resulting from rail impact testing in accordance with 4.5.2.2.4 without failure, damage, or permanent deformation.

3.7.5 Highway transportability. The press when loaded on a semitrailer/tractor, shall be within the highway permit limits of all States.

3.8 Human factors engineering. The press shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), control/display integration (5.1), visual displays (5.2), controls (5.4), labeling (5.5), anthropometry (5.6), design for maintainability (5.9), hazards and safety (5.13) of MIL-STD-1472.

3.9 Noise limits. The noise produced by the press and associated components shall conform with MIL-STD-1474 requirements, with the exception of

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MIL-STD-1474 (5.2, 5.3, and 5.4), when tested in accordance with 4.5.2.4. The provisions of MIL-STD-1474 (4.2 and 4.3), shall be provided if and only if MIL-STD-1474 (5.1.1.2), procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the procuring activity. Hazard signs shall conform with MIL-STD-1474, 4.2.

3.10 Safety. All rotating, reciprocating, high temperature and projecting parts so located as to be hazardous to operating personnel shall be fully guarded, enclosed or insulated to the extent necessary to eliminate the hazard. Emergency press-stop buttons shall be of a distinctive shape and color and so located as to be accessible and identifiable from all operating locations. A safety bar shall be located at the gap between the plate and the blanket cylinders to protect the operator. Actuation or movement of this bar in either direction will immediately stop the press.

3.11 Storage temperature. The press shall not be damaged by storage in any ambient temperature from +160 °F (+71 °C) to -30 °F (-34 °C). Damage shall include malfunction, breakage, inability to operate or permanent deformation.

3.12 Capacities and dimensions. The press shall have the following capacities and dimensions:

Technical data

Maximum sheet size	24.00 x 32.25 in. (610 x 820 mm)
Minimum sheet size	11.00 x 16.50 in. (280 x 420 mm)
Maximum sheet thickness	6-ply 0.032 in. (0.813 mm)
Minimum sheet thickness	9-pound onionskin 0.002 in.(0.051 mm)
Maximum image area	23.38 x 32.25 in. (595 x 820 mm)
Gripper margin	0.31 to 0.38 in. (8-10 mm)
Maximum printing speed	10,000 sheets per hour (sph)
Minimum printing speed	2,000 sheets per hour (sph)
Cylinder diameter	10.630 in. (270 mm)

Plate cylinder

Length, normal/maximum	27.25/28.12 in. (692/715 mm)
Width	32.50 in. (825 mm)
Thickness	0.006-0.020 in. (0.15-0.50 mm)
Cylinder undercut	0.020 in. (0.5 mm)
Distance from plate edge to begin print	2.19 in. (55.5 mm)

Blanket cylinder

Length	27.16 in. (690 mm)
Width	33.53 in. (852 mm)

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Thickness	0.075 in. (1.9 mm)
Cylinder undercut	0.126/0.090 in. (3.2 mm [2.3])
Inking unit	
Form inking rollers	4
Diameter of form inking rollers	2.460 in. (62.5 mm), 2.244 in. (57 mm), 2.362 in. (60 mm), 2.559 in. (65 mm)
Height of piles (from table top to top of pile)	
Feeder	39.38/42.91 in. (1000 mm/1090)
Delivery	17.72/22.84 in. (450/580 mm)
Floor space, maximum	115 x 70 in. (2.92 x 1.77 m)
Overall height, maximum	77.38 in. (1.84 m)
Net weight, maximum	12,040 lb. (5,466.2 kg)

3.12.1 Mechanical run. When tested as specified in 4.5.2.7, the press shall be capable of feeding through the maximum sheet size paper without any interruption of feeding, stock jamming, gripper nicks or other paper markings.

3.12.2 Paper capacity and feed. When tested as specified in 4.5.2.8, the press shall be capable of feeding through the full range of sheet thicknesses of the maximum sheet size paper at maximum printing speed without any interruption of feeding, stock jamming, gripper nicks or other paper markings.

3.12.3 Size and register. When tested as specified in 4.5.2.9, the printed paper shall show no evidence of doubling or elongation of dots or lines on any of the sheets when examined under a 9-power magnifying glass.

3.12.4 Ink coverage. When tested as specified in 4.5.2.10, the printed paper shall show no evidence of increases in dot size or filling-in shadow detail when examined under a 9-power magnifying glass, uneven ink distribution, or gray areas in the solid background.

3.13 Press base and side frames. The press base and side frames shall be steel or cast iron and shall form a rigid, nondistorting, supporting structure for the press components. All mounting surfaces shall be machined and all bearings and bushings shall be aligned to maintain parallelism and contact between all rotating cylinders and rollers.

3.14 Feeder and suction head. The feeder and suction head shall be automatic, center-separation, suction-stream-fed type with provisions for automatically raising the paper pile, separating the upper sheets, and

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delivering them in accurate alignment and sequence over the feedboard to the sheet-registering mechanism. A motorized powerhoist shall be incorporated for independently raising and lowering the paper pile.

3.15 Sheet-registering mechanism.

3.15.1 Guides. The registering mechanism shall have provisions for both right- and left-side guiding. The side guides shall be adjustable for any width of sheet within the rated capacity of the press and shall be the pull type. A micrometer adjustment for fine register shall be an integral part of both side guides and shall be designed to permit adjustments while printing. The registering mechanism shall incorporate an adjustable, multiple-sheet choke to prevent more than one sheet at a time from passing through to the front guides. An electronic detector shall be incorporated to detect if a sheet arrives too early, too late, or askew of the front guides. When activated, the detector shall automatically trip the press off impression, lift the ink-form rollers from the press plate, feeder and press shall stop altogether.

3.15.2 Register mechanism. After the sheet is side registered, the sheet shall be overfed by a vacuum device to the adjustable paper stops so the impression cylinder grippers may properly close on the paper.

3.16 Press cylinders. Press cylinders shall be steel weldments or steel or iron castings which have been stress-relieved. The plate and blanket cylinders shall have figures stamped on the gap edge to indicate the depth to which the cylinders are undercut below the bearer height. All cylinders shall be balanced and supported on antifriction components to provide smooth-running vibration-free operation. All press cylinder gears shall be furnished with hardened and accurately ground or shaved tooth contour with accurate involute control to preclude gear streaks, slurs, or other printing distortions.

3.16.1 Plate cylinder. The plate cylinder circumference shall be undercut below its bearer height by the amount specified in 3.12. The plate cylinder shall be equipped with adjustable, quick-acting plate clamps for securing the plate around the cylinder with a minimum of edge bending and distortion. The plate clamps shall be of adequate capacity and lead to accommodate new and reused plates and to hold them securely in position for the duration of the press run. The plate clamps shall shift and swing the plate at least 0.125 inch (3.175 mm) in any direction as may be required for registration. The plate cylinder shall also be adjustably secured to its gear with provisions for full 360 degrees micrometric advancing or retarding its relative surface position. The plate-cylinder gear shall consist of a paired gear assembly with internal provisions for a controlled displacement to take up gear backlash. When specified (see 6.2), the front plate clamp strip shall be equipped for utilizing a pre-register system.

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3.16.2 Blanket cylinder. The blanket cylinder shall be undercut below its bearer height by the amount specified in 3.12. The cylinder shall incorporate facilities for clamping and uniformly stretching and holding the rubber blanket around its circumference. The cylinder shall be mounted and controlled to be automatically displaced out of printing position when the sheet detectors show that a sheet of paper is not in correct position for printing. To compensate for printing pressures, the cylinder shall be adjustably controlled from the operator's side of the press to correspond with the range of paper thickness specified in 3.12. The cylinder shall maintain its parallelism to the plate and impression cylinders when adjusted for any printing pressure required.

3.16.3 Impression cylinder. The impression cylinder shall be undercut below its gear pitch diameter to produce dimensional reproduction on offset book paper. The cylinder shall incorporate positive cam action closing of the paper grippers to hold the sheet in register for printing and to release it for acceptance by the press chain delivery.

3.17 Ink fountain. The ink fountain shall be adjustable to control the ink supply across the width of the printing plate and to control the amount of ink transferred to the ductor roller. The ink fountain keys shall be number coded for fast adjustment to light and heavy printing areas. The ink fountain shall also incorporate an arrangement for moving the ink fountain blade away from the fountain roll to facilitate cleaning the blade. The method used shall be such that the ink fountain blade will be returned to its original operating position without further adjustment to the blade adjusting screws.

3.18 Ink distribution. The ink distribution system shall have the minimum numbers of rollers which will accept, distribute, and supply a uniform quantity of ink across the width of the form rollers. The system shall ink the printing area of the press plate solidly without holidays. It shall also ink any combination of solids and halftones without filling in halftones or increasing the dot sizes.

3.19 Washup attachment. Each press shall be equipped with a washup attachment for removing the ink from the entire distribution system in cleaning the press with the aid of ink solvents. The washup attachment blade shall be replaceable.

3.20 Dampening system. The dampening system shall utilize a non-hazardous dampening fluid in a motorized circulation system. An integral system of distributing rollers shall deliver a controlled quantity of moisture as required for a correct ink-moisture balance on various types of plates and surfaces. The drive of the dampening system shall be independent from the drive for more accurate control of ink-moisture balance. Contact of dissimilar metals throughout the water distribution system shall be avoided. Chrome plating, when used, shall be controlled to provide the porosity required for uniform water acceptance and continued service in the dampening system of the lithographic press.

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3.21 Press delivery. The press shall incorporate a chain-driven series of gripper bars with individual spring-loaded paper grippers and actuating mechanism to grasp the sheets in sequence as they are released by the impression cylinder grippers. The gripper bars shall carry the sheets to the delivery pile, releasing the sheets at a predetermined adjustable release point for automatic jogging and stacking on the receding pile delivery. The press delivery shall incorporate uninterrupted unloading facilities to provide continuous delivery during the press run and shall be equipped with vacuum wheels and airbars for positive sheet handling and even sheet stacking. The delivery shall have a fluorescent light over the pile and be equipped with a transparent guard to permit accurate inspection of the printed sheets by the press operator.

3.22 Electrical requirements. Unless otherwise specified (see 6.2), the press shall operate on 200-230/460 volt, 3 phase, 60 Hertz alternating-current, grounded wye power supply. All electrical equipment and wiring shall conform to NFPA No. 70.

3.22.1 Electrical-electronic equipment. All electrical-electronic equipment shall be furnished fully wired to terminal boxes with marked terminals and correspondingly marked interconnecting cable or wiring assemblies where necessary. Press operating controls shall consist of "start", "stop", and "inch" at feeder and delivery locations plus "reverse" at delivery end of press with additional controls where required to facilitate operation. All stop buttons shall have a safety catch or ring to prevent operation of the press when the safety catch is latched. The control buttons shall be so wired that the run and inch buttons must be depressed simultaneously to start the press in continuous operation. A trip switch shall be coupled directly to the impression lever to automatically reduce press speed from its running speed to its idling speed when the printing pressure is released. Limit switches to prevent overtravel of the pile-feeder drive shall be provided.

3.22.2 Motors. Motors shall conform to NEMA MG1, type and size as applicable.

3.23 Fungus and moisture resistance. When specified (see 6.2),the electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.23.1), with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.

b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

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When used, the varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element. Surfaces shall be previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.23.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition II is required, and shall certify that the composition II material complies with Rule 442, South Coast Air Quality Management Control District.

3.24 Lubrication. All surfaces requiring lubrication shall be provided with a means for automatic or manual application.

3.24.1 Lubricants. The press shall operate as specified herein when lubricated with military lubricants as specified in MIL-L-2104, grade 10, and MIL-G-25013.

3.24.2 Lubrication fittings. Lubrication fittings shall conform to MS15003. Fittings shall be located in a protected position and shall be accessible to a grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility to lubrication fittings. Areas to be lubricated manually shall be color coded to indicate the correct time interval for lubrication.

3.24.3 Delivery lubrication service. When delivered, all components except those for which preservatives are specified shall be lubricated. Lubrication shall be with oils or greases conforming to MIL-L-2104 or MIL-G-25013, as applicable. Special lubrication instruction plates shall be furnished in a conspicuous place to indicate the temperature range and grades of oils and greases used.

3.25 Identification marking. The press shall be identified in accordance with MIL-STD-130. The marking shall be applied on plates conforming to MIL-P-514, type III, composition C (GG-P-455, type optional, grade B, class 1). The plate shall be attached to the press by screws, bolts, or rivets in a place where the plate will be both visible and legible but not interfere with the operation and maintenance of the press.

3.26 Instruction plates or labels. Each press shall be equipped with instruction plates, labels, or diagrams, including warnings and cautions, describing any special or important procedures to be followed in assembling, operating, or servicing the press. Instruction plates and labels shall conform to MIL-P-514, type III, composition C (GG-P-455, type optional, grade B, class 1), and the labeling requirements of 3.8. Plates or labels shall be

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attached by screws, bolts, or rivets in a place where the plates or labels will be both visible and legible but not interfere with the operation and maintenance of the press.

3.27 Treatment and painting. The portions of the press normally painted shall be cleaned, treated, and painted in accordance with the contractor's usual standard practice.

3.28 Workmanship. All parts, components and assemblies of the press, including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts, shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operation, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article press shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article press shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Examination. Each press shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests. Each press shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The press shall be examined as specified herein for the following defects:

101. Materials not as specified (see 3.3).
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration of the applicable operating and storage environments (see 3.3.1).
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of materials, material finishes, or treatment (see 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces, and parts incorporated in the press (see 3.3.2).
106. Press not of manufacturer's product lines as specified (see 3.4).
107. Design not as specified (see 3.5).
108. Transportability provisions not as specified (see 3.7.1 thru 3.7.5).
109. Capacities and dimensions not as specified (see 3.12).
110. Press base and side frames not manufactured as specified (see 3.13).
111. Feeder and suction head not as specified (see 3.14).
112. Guides not as specified (see 3.15.1).
113. Register mechanism not as specified (see 3.15.2).
114. Press cylinders not as specified (see 3.16).
115. Plate cylinder not as specified (see 3.16.1).
116. Blanket cylinder not as specified (see 3.16.2).
117. Impression cylinder not as specified (see 3.16.3).
118. Ink fountain not as specified (see 3.17).
119. Ink distribution not as specified (see 3.18).

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120. Washup attachment not as specified (see 3.19).
121. Damping system not as specified (see 3.20).
122. Press delivery not as specified (see 3.21).
123. Electrical requirements not as specified (see 3.22).
124. Electrical-electronic equipment not wired or connected as specified (see 3.22.1).
125. Motor not as specified (see 3.22.2).
126. Fungus and moisture resistance treatment not as specified (see 3.23).
127. All surfaces requiring lubrication not provided with a means for lubricating (see 3.24).
128. Lubricants not as specified (see 3.24.1).
129. Lubrication fittings not as specified (3.24.2).
130. Delivery lubrication service not as specified (see 3.24.3).
131. Identification marking not as specified (see 3.25).
132. Instruction plates or labels not as specified (see 3.26).
133. Treatment and painting not as specified (see 3.27).
134. Workmanship not as specified (see 3.28).

4.5.2 Tests. Prior to test, the press shall be lubricated with oils and greases conforming to MIL-L-2104, grade 10, and MIL-G-25013.

4.5.2.1 Ease of maintenance. The press shall be evaluated for maintenance accessibility throughout testing. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.2 Transportability.

4.5.2.2.1 Tiedown provisions. The tiedown provisions shall be tested in accordance with MIL-STD-209 to provide conformance to 3.7.1. Inability to meet the requirements of 3.7.1 shall constitute failure of this test.

4.5.2.2.2 Slinging provisions. The slinging provisions shall be tested in accordance with MIL-STD-209 to prove conformance to 3.7.2. Inability to meet the requirements of 3.7.2 shall constitute failure of this test.

4.5.2.2.3 Air transport. The press shall be tested in accordance with MIL-STD-1791 to prove conformance to 3.7.3. Inability to meet the requirements of 3.7.3 shall constitute failure of this test.

4.5.2.2.4 Rail impact. The first article press shall be tested in accordance with MIL-STD-810 to prove conformance to 3.7.4. Inability to meet the requirements of 3.7.4 shall constitute failure of this test.

4.5.2.3 Human factors engineering. Human factors engineering requirements shall be inspected throughout testing. Nonconformance to 3.8 shall constitute failure of this test.

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4.5.2.4 Noise level. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 11. The press shall be tested inside an International Standardization Organization (ISO) container (van body) with air conditioning, heating, and utilities. As a minimum, noise levels shall be measured when equipment is operating at maximum capacity. The noise level at the operator's typical operating position shall be provided as dB(A) level. Failure to comply with 3.9 and MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.5 Safety. The press shall be evaluated for safety requirements throughout testing. Nonconformance to 3.10 shall constitute failure of this test.

4.5.2.6 Storage temperature. The press shall be placed in an environmental test chamber. The internal chamber temperature shall be lowered to -30 ±5 °F (-34 °C) and maintained at that temperature for not less than 1 hour. Then the chamber temperature shall be raised to +160 ±5 °F (+71 °C) and maintained for not less than 5 hours at that temperature. After the chamber temperature has returned to ambient conditions, the press shall be examined and tested in accordance with 4.5.2.7. Nonconformance to 3.11 and the failure criteria of 4.5.2.7 shall constitute failure of this test.

4.5.2.7 Mechanical run. The feeder platform shall be stacked with a full load of 24.00 by 32.25-inch (610 by 820 mm) paper equivalent in weight to high, wet-strength paper conforming to MIL-P-43027, type I. All necessary press adjustments shall be made for this stock. The press shall be inked with ink conforming to MIL-I-3606, color optional. The press shall be operated continuously, with paper feeding but without printing, at the maximum rate of speed until the entire load is fed through. Slowdowns necessary to remove full-delivery piles shall be permitted. Nonconformance to 3.12.1 shall constitute failure of this test.

4.5.2.8 Paper capacity and feed. The press shall be tested with the full range of paper stock from 9-pound onionskin to 6-ply card 24.00 by 32.25 inches (610 by 820 mm) in size. Individual quantities of at least 50 sheets of each size and type shall be fed through the press (impression "on" but without printing) at 2,000 sheets per hour (sph) (except that the paper comparable to high wet-strength paper shall be run at 10,000 sph). Nonconformance to 3.12.2 shall constitute failure of this test.

4.5.2.9 Size and register. A press plate containing various line and halftone subjects (120-line screen or finer) covering the rated area, with hairline (approximately 0.003 inch (0.076 mm) in width) corner marks shall be printed on 24.00 by 32.25-inch (610 by 820 mm) paper at a speed of 10,000 sph. After not less than 75 sheets have been fed through the press, they shall be run through the press again so that the second impression will print on the first. Nonconformance to 3.12.3 shall constitute failure of this test.

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4.5.2.10 Ink coverage. A press plate, containing at least six assorted small halftone subjects, each approximately 2 by 3 inches (50.8 by 76.2 mm) in size (screened 120 lines or finer) distributed over the printing area, with the remainder of the printing area of the plate filled in solid, shall be printed in full-color density on at least 75 sheets of 24.00 by 32.25-inch (610 by 820 mm) paper. Nonconformance to 3.12.4 and 3.18 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.2 Unit of product. For the purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.6.3 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.4 Examination. Samples selected in accordance with 4.6.3 shall be examined for applicable defects listed in table I. One or more defects shall be cause for rejection.

TABLE I. Examination.

Number	Characteristics	Requirement Paragraph
135.	Disassembly not as specified for level A or B.	5.2.1.1, 5.2.2.1
136.	Matchmarking not as specified for level A or B.	5.2.1.2, 5.2.2.1
137.	Tags used for matchmarking not waterproofed for level A.	5.2.1.2
138.	Exterior surfaces not coated with preservative as specified for level A or B.	5.2.1.4, 5.2.1.5, 5.2.2.3, 5.2.2.4
139.	Unpainted metal information plates not coated with varnish for level A.	5.2.1.4
140.	Enclosed gears not preserved and tagged as specified for level A or B.	5.2.1.6, 5.2.2.5
141.	Enclosed drive chains not preserved and tagged as specified for level A or B.	5.2.1.7.1, 5.2.2.6.1
142.	Exposed chains not preserved as specified for level A or B.	5.2.1.7.2, 5.2.2.6.2
143.	Removed chains not coiled and wrapped as specified for level A or B.	5.2.1.7.2, 5.2.2.6.2
144.	Disk-type clutches not preserved as specified for level A.	5.2.1.8.1

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TABLE I. Examination. (Con't)

Number	Characteristics	Requirement Paragraph
145.	Disk-type clutches not disengaged as specified for level A or B.	5.2.1.8.1, 5.2.2.7
146.	Band-type clutches not preserved as specified for level A.	5.2.1.3.2
147.	Drive belts not removed or released from tension for level A or B.	5.2.1.9, 5.2.2.8
148.	Pulley faces or grooves not coated with primer as specified for level A.	5.2.1.9
149.	Belts removed not preserved as specified for level A or B.	5.2.1.9, 5.2.2.8
150.	Electric motors not preserved as specified for level A or B.	5.2.1.10, 5.2.2.9
151.	Hydraulic system not preserved as specified for level A or B.	5.2.1.11, 5.2.2.10
152.	Hydraulic control valves not secured in neutral position for level A or B.	5.2.1.11, 5.2.2.10
153.	Ends of wires, electrical tables, plug openings, terminals, openings into switches junction boxes, and couplings not covered or sealed as specified for level A.	5.2.1.12
154.	Openings into gages not sealed with tape as specified for level A.	5.2.1.13
155.	Cages and indicators removed not preserved as specified and threads and threaded openings not coated as specified for level A or B.	5.2.1.13, 5.2.2.11
156.	Centrifugal pumps not preserved in accordance with the referenced document for level A or B.	5.2.1.14, 5.2.2.12
157.	Hand pumps not preserved and the openings sealed as specified for level A or B.	5.2.1.15, 5.3.2.13
158.	Air compressors not preserved in accordance with the referenced document for level A or B.	5.2.1.16, 5.2.2.14
159.	Ends of pipe and tubing not sealed as specified for level A.	5.2.1.17
160.	Loose valves, fittings, and flanges not preserved in accordance with referenced document for level A.	5.2.1.18

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TABLE I. Examination. (Con't)

Number	Characteristics	Requirement Paragraph
161.	Float valve mechanisms not secured and tagged as specified for level A or B.	5.2.1.18, 5.2.2.15
162.	Hose and hose fittings not preserved in accordance with the referenced document for level A.	5.2.1.19
163.	Openings into ends of hose not removed and fittings not sealed as specified for level A.	5.2.1.19
164.	Lithographic presses, not preserved as specified for level A or B.	5.2.1.20, 5.2.1.20.1, 5.2.1.20.2,. 5.2.2.16
165.	Repair parts not preserved as specified for level A or B.	5.2.1.21, 5.2.2.17
166.	Maintenance tools not preserved as specified for level A.	5.2.1.22
167.	Technical publications not preserved as specified for level A.	5.2.1.23
168.	Consolidation not as specified for level A or B.	5.2.1.24.1, 5.2.2.18
169.	Packing container not as specified (class, style, type) for level A or B.	5.3.1, 5.3.2
170.	Boxes not modified with skids for level A or B.	5.3.1, 5.3.2
171.	Strapping not zinc coated for level A.	5.3.1
172.	Marking illegible, incomplete, incorrect or missing for level A, B or C.	5.4

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination and tests within the time frame specified (see 6.2) to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packaging requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The first article pack may be accomplished utilizing either the first article press or a production press. If the first article press is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article press and production press.

5.2 Preservation. Preservation shall be level A, B, or C, as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Disassembly. Disassembly of the units of equipment shall be the minimum necessary to safeguard parts vulnerable to damage, pilferage and loss. The extent of further disassembly shall be confined to the removal of protruding parts that would otherwise increase cubage. However, disassembly shall be limited to those parts that can be removed and reinstalled without special skill and tools. Removed bolts, nuts, pins, screws, and washers shall be reinstalled in mating parts and secured to prevent their loss.

5.2.1.2 Matchmarking. Removed parts and mating parts on the units of equipment shall be matchmarked when necessary to facilitate reassembly. Removed parts shall be identified by cloth shipping tags conforming to UU-T-81, type A, and the marked tags shall be attached to mating parts. The tags and the printing thereon shall be waterproofed in accordance with MIL-STD-129.

5.2.1.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in, and shall be applied in accordance with MIL-P-116.

5.2.1.4 Threaded and unpainted exterior surfaces. Threaded and unpainted exterior ferrous metal surfaces of items such as springs, stands, bases, pins, linkages, grease fittings, bolts and nuts, or any other unpainted metal surface not specifically identified herein, requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1, preservative. Unpainted ferrous metal information plates shall be coated with varnish conforming to MIL-V-13811.

5.2.1.5 Exposed machined metal surfaces. Exposed machined ferrous metal surfaces such as shafts, worm gears, levers, and working parts of the equipment shall be coated with type P-6 or P-11 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wrap or cover shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.6 Enclosed gears. Gears operating in engine lubrication oil (grade 10, 30, or 50) shall have the housing drained and filled to the operating level with type P-10, grade 1, 2, or 3 preservative, as applicable, and tagged to indicate the preservative oil is suitable for operation and shall remain in the housing until the first required lubricant change. Gears operating in other than engine lubricating oil shall have the housing filled to the operating level with the approved lubricant required by the lubrication order for operation and tagged to indicate the housing is filled with the required operating oil.

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5.2.1.7 Drive chains.

5.2.1.7.1 Enclosed chains. Enclosed drive chains and chain housings shall be preserved and tagged as specified in 5.2.1.6 for enclosed gears.

5.2.1.7.2 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. The excess preservative shall be drained. Chains, if removed, shall be preserved as specified herein and the chains shall be individually coiled. Each coil shall be secured with soft annealed wire and each coil shall be wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The wraps shall be secured with tape conforming to PPP-T-60, type IV.

5.2.1.8 Clutches.

5.2.1.8.1 Disk-type clutches. With the cover plates removed and the clutch engaged, all accessible metal interior components of the clutch shall be sprayed with a thin film of primer conforming to MIL-P-53030. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After being sprayed, spring-loaded-type clutches shall be secured in a partially disengaged position to eliminate contact between the disk facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

5.2.1.8.2 Band type. Clutch drum facings shall be coated with primer conforming to MIL-P-53030. Mechanism enclosed within the housing shall be coated with type P-1 preservative.

5.2.1.9 Drive belts and pulleys. Drive belts shall be removed or released from tension. Pulley faces or grooves shall be coated with primer conforming to MIL-P-53030. Removed belts shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.10 Electric motors. Electric motors, except those attached to units preserved in accordance with MIL-P-116, method II, shall be preserved in accordance with MIL-E-16298, level A, using the alternate method specified therein for assembled machines.

5.2.1.11 Hydraulic systems. The hydraulic fluid supply tanks shall be filled with the hydraulic fluid recommended for operation. The pistons shall be fully retracted into the cylinders and secured. Any exposed portions of the piston rods (ramshafts) shall be coated with type P-6 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to PPP-T-60, type IV. The hydraulic control valves shall be secured in the neutral position. Hose shall not be disconnected. Any unpainted exterior

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metal surfaces of the system requiring the application of a contact preservative in accordance with MIL-P-110 shall be coated with type P-1 preservative.

5.2.1.12 Controls and wiring. Except for equipment preserved in accordance with MIL-P-116, method II, exposed ends of wires, electrical cables, plug openings, terminals, and openings into switches, junction boxes, and couplings shall be covered with caps or plugs conforming to MIL-C-5501, or sealed with tape conforming to PPP-T-60, type IV.

5.2.1.13 Gages and indicators. Except for equipment preserved in accordance with MIL-P-116, method II, all openings into the interior of gages and indicators mounted on the equipment shall be sealed with caps or plugs conforming to MIL-C-5501, or tape conforming to PPP-T-60, type IV. The faces of dial glasses shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and the barrier material secured in place with tape specified herein. Gages and indicators removed from the equipment, including equipment preserved in accordance with MIL-P-116, method II, shall be individually preserved in accordance with MIL-P-116, method IC-2. The threads of openings exposed by removal and those on the gage or indicator shall be coated with type P-10, grade 1 preservative.

5.2.1.14 Centrifugal. Centrifugal pumps and pump components shall be preserved in accordance with MIL-P-10603, level A.

5.2.1.15 Hand pumps. The pump check valve shall be removed, and the interior of the pump shall be flushed with type P-10, grade 2 preservative. The pump shall be actuated to assure coating of all interior surfaces. Excess preservative shall be drained. Drain-plug threads shall be coated with preservative specified herein and the drain plugs and check valve shall be reinstalled. Openings into the pump shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and the barrier material shall be secured in place with tape conforming to PPP-T-60, type IV.

5.2.1.16 Air compressors. Air compressors and their components shall be preserved in accordance with MIL-C-3600, level A.

5.2.1.17 Pipe and tubing. The ends of loose pipe and tubing shall be sealed with plastic caps or plugs conforming to MIL-C-5501, or with tape conforming to PPP-T-60, type IV.

5.2.1.18 Loose valves, fittings, and flanges. Loose valves, fittings, and flanges shall be preserved in accordance with MIL-V-3, level A. In addition, all valves with float mechanisms shall have the floats secured against movement. A waterproof tag shall be attached to the valve stating: "Float has been secured and must be released before valve is placed in operation."

5.2.1.19 Hose and hose fittings. Hose and hose fittings removed from the unit shall be preserved in accordance with MIL-H-775, level A. Openings into

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ends of hose not removed and fittings not removed, except openings on units preserved in accordance with MIL-P-116, method IIa, shall be sealed with plastic caps or plugs conforming to MIL-C-5501 or with tape conforming to PPP-T-60, type IV. Hose on reels shall be secured to prevent unwinding.

5.2.1.20 Lithographic presses. Each lithographic press shall be preserved in accordance with MIL-P-116, method IIa. The press shall be secured directly through the barrier material to the base of the shipping container specified in 3.4.

5.2.1.20.1 Ink, dampening, and ductor rollers. The ink, dampening, and ductor rollers shall be removed. The metal ends of the rollers shall be coated with type P-1 preservative. Each roller shall be wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2. The wraps shall be secured with tape conforming to PPP-T-60, type IV. The ends of the rollers shall be supported with wood blocking in a manner that the rolls will not flatten when placed in a container for consolidation.

5.2.1.20.2 Press cylinders and steel rollers. The press cylinders and steel rollers shall be coated with type P-6 preservative. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, and the wrap or cover shall be secured in place with tape conforming to PPP-T-60, type IV. The press cylinders shall be adjusted so that the bearing surfaces shall not be in contact, when on impression. The rollers shall be secured to prevent movement.

5.2.1.21 Repair parts. Unless otherwise specified (see 6.2), the preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.22 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.23 Technical publications. Technical publications for each basic unit shall be preserved together in accordance with MIL-P-116, method IC-1.

5.2.1.24 Consolidation.

5.2.1.24.1 Equipment with chests. When a chest is furnished to contain the equipment, the equipment components, accessories, repair parts, tools, and technical publications comprising the complete item or equipment shall be consolidated in the chest. Cushioning and blocking shall be provided to prevent movement or damage to the items in the chest. The chest lid shall be closed and secured.

5.2.1.24.2 Components, accessories, repair parts, tools, and publications for equipment without chests. The components, accessories, repair parts, tools, and publications for equipment without chests shall be consolidated as follows: For equipment to be packed in nailed wood boxes the components

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accessories, repair parts, tools, and publications for the equipment shall be consolidated in boxes conforming to PPP-B-636, W5c, style optional. For equipment to be packed in crates, the components, accessories, repair parts, tools, and publications for the equipment shall be consolidated in boxes conforming to PPP-B-621, class 1, style optional; or in boxes conforming to PPP-B-601, domestic type, style optional. The boxes shall be close fitting. Strapping of the boxes shall not be required.

5.2.2 Level B.

5.2.2.1 Disassembled and matchmarkings. Disassembly and matchmarking shall be as specified for level A, except waterproofing of cloth tags shall not be required.

5.2.2.2 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116.

5.2.2.3 Threaded and unpainted exterior surfaces. Threaded and unpainted exterior surfaces shall be coated with type P-1 preservative as specified for level A.

5.2.2.4 Exposed machined metal surfaces. Exposed machined ferrous metal surfaces shall be coated with type P-6 preservative and wrapped or covered as specified for level A.

5.2.2.5 Enclosed gears. Gear housing shall be filled to the operating level with the approved lubricant required for operation. The gears shall be operated to insure coating of all interior surfaces with the lubricant. The unit shall be tagged to indicate the housing is filled with the required amount of operating oil.

5.2.2.6 Chains.

5.2.2.6.1 Enclosed chains. Enclosed chains and chain housings shall be filled with operational lubricant, operated, and tagged as specified in 5.2.2.5 for enclosed gears.

5.2.2.6.2 Exposed chains. Exposed drive chains shall be coated with enough type P-3 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. Chains, if removed, shall be preserved as specified herein, coiled, the coil secured, and each coil wrapped with barrier material conforming to MIL-B-121, type II, grade A, class 2. The wrap shall be secured with tape conforming to PPP-T-60, type IV.

5.2.2.7 Disk-type clutches. Spring-loaded-type clutches shall be secured in a partially disengaged position to eliminate contact between the disk facings and pressure plates. Snap-over-center and toggle-in-type clutches shall be completely disengaged.

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5.2.2.8 Drive belts. Drive belts shall be released from tension. Belts, if removed, shall be preserved in accordance with MIL-P-116, method III.

5.2.2.9 Electric motors. Electric motors, except those attached to equipment preserved by method IA-16 in accordance with MIL-P-116, shall be preserved in accordance with the level A requirements of MIL-E-16298, using the alternate method specified therein for assembled machines.

5.2.2.10 Hydraulic systems. Hydraulic systems shall be preserved as specified for level A.

5.2.2.11 Gages and indicators. Gages and indicators removed from the equipment, including equipment preserved by method IA-16 shall be individually preserved in accordance with MIL-P-116, method III. The threads of openings exposed by removal and those on the gage or indicators shall be coated with type P-10, grade 1 preservative.

5.2.2.12 Centrifugal pumps. Centrifugal pumps and pump components shall be preserved as specified for level A.

5.2.2.13 Hand pumps. Hand pumps shall be preserved as specified for level A.

5.2.2.14 Air compressors. Air compressors and their components shall be preserved in accordance with MIL-C-3600, level B.

5.2.2.15 Float valve mechanisms. All float valve mechanisms shall be secured to prevent movement and tagged to indicate: "Float has been secured and must be released before the valve is placed in operation."

5.2.2.16 Lithographic presses, printing and dry developing machines, blue print machines, duplicating machines, plate coating machines, phototypesetters and contact printer assemblies. The machines, components and accessories shall be preserved as specified for level A, except that when method IIa is specified, method IA-16, in accordance with MIL-P-116, shall be substituted.

5.2.2.17 Repair parts, maintenance tools, and technical publications. Repair parts, maintenance tools, and technical publications shall be preserved as specified for level A.

5.2.2.18 Consolidation. Consolidation shall be as specified for level A, except fiber boxes shall be domestic class, SW variety.

5.2.3 Level C. Each complete press shall be preserved in a manner to afford protection against deterioration and damage during shipment from the contractor to the initial destination.

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5.3 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.3.1 Level A. Each complete unit of equipment, with the required components, supplies, accessories, repair parts, tools, and publications (including equipment in chests), not to exceed the weight limitation of the box specification, shall be packed as a complete unit in a box conforming to PPP-B-621, class 2, style 2, 2-1/2, 3, or 7; or in a box conforming to PPP-B-601, overseas type, style optional. The boxes shall be modified with skids in accordance with the box specification, and when the height of the boxes exceeds 36 inches, diagonals shall be used on style 2, 2-1/2, and 3 applied as specified in PPP-B-621 for type 3 loads. The boxes shall be close fitting. Strapping shall be in accordance with the appendix to the box specification, zinc-coated and placed adjacent to the skids on modified boxes. Each complete unit of equipment, with the required components, supplies, accessories, repair parts, tools, and publications (including equipment in chests), exceeding the weight limitation of a nailed wood box shall be packed in a crate conforming to MIL-C-104, type I, class 1 or 2, style A. Blocking, bracing, anchoring, closure, and strapping shall be in accordance with appendix to the crate specification and in accordance with MIL-STD-1186. Strapping shall be zinc coated.

5.3.2 Level B. Each complete unit of equipment, with the required components, supplies, accessories, repair parts, tools, and publications (including equipment in chests), not to exceed the weight limitation of the box specification, shall be packed as a complete unit in a box conforming to PPP-B-621, class 1, style optional, or in a box conforming to PPP-B-601, domestic type, style optional. The boxes shall be modified with skids in accordance with the box specification. When the height of a box exceeds 36 inches, diagonals shall be used on boxes conforming to PPP-B-621. The diagonals shall be applied as specified for type B loads. The boxes shall be close fitting and shall be strapped in accordance with the appendix to the applicable box specifications, except straps shall be placed adjacent to the skids on modified boxes. Each complete unit of equipment, with the required components, supplies, accessories, repair parts, tools, and publications (including equipment in chests), exceeding the weight limitation of the wood boxes shall be packed in a crate conforming to MIL-C-52950, type V, style A, class 1, or MIL-C-3774, nailed assembly. Blocking, bracing, anchoring, waterproofing of contents, closure, and strapping shall be in accordance with the crate specification and appendix thereto and in accordance with MIL-STD-1186.

5.3.3 Level C. Each complete press, preserved as specified in 5.2, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.4 Marking. In addition to any special marking specified by the contract or purchase order (see 6.2), marking shall be in accordance with MIL-STD-129.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The press is intended to print military maps, photomaps, and other data; primarily in the mobile configuration but can be used also in a fixed installation.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2, 4.3 and 6.4).
d. When air transportability data is required (see 3.7.3).
e. When pre-register system is required (see 3.16.1).
f. Electrical requirements if other than specified (see 3.22).
g. When fungus and moisture resistance treatment is required (see 3.23).
h. Time frame required for submission of first article pack (see 5.1 and 6.5).
i. Level of preservation and packing required (see 5.2 and 5.3).
j. Specific preservation instructions for repair parts when other than as specified (see 5.2.1.2.1)
k. Any special marking (see 5.4).
l. When a transportability report is required (see 6.7).

6.3 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.4 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior

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Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 First article pack. Any changes or deviations of production packs from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the press in accordance with this specification.

6.6 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of printing presses.

6.7 Transportability report. When specified (see 6.2), the contracting officer should arrange for the contractor to furnish a transportability report in accordance with AR 70-47 and S.O.P 70-38 for the press, 95 days prior to delivery of the first article press.

6.8 International standardization agreements. Certain provisions of this specification are the subject of international standardization agreement QSTAG 111. When amendment, revisions, or cancellation of this specification is proposed that will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels, including departmental standardization offices, to change the agreement or make other appropriate accommodations.

6.9 Subject term (key word) listing.

Blanket cylinder
Dampening system
Feeder
Guides
Impression cylinder
Ink distribution
Ink fountain
Motors
Plate cylinder
Press base
Press cylinders
Map printing
Press delivery
Register mechanism
Side frames
Suction head
Washup attachment

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6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - ME

Preparing activity:
Army - ME

Review activities:
Army - EL, GL
DLA - GS

Project 3610-A371

Coordinating activity:
GSA-FSS-FCGC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-52434C(ME)	2. DOCUMENT DATE (YYMMDD) 92/01/30

3. DOCUMENT TITLE
Printing Press, Offset, Lithographic (Motor Driven) 24 X 30 Inches Maximum Sheet Size

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (if applicable) (2) AUTOVON	7. DATE SUBMITTED

8. PREPARING ACTIVITY

a. NAME Violet Stewart	b. TELEPHONE (Include Area Code) (1) Commercial (703) 664-1806	(2) AUTOVON 354-1806
c. ADDRESS (Include Zip Code US Army Belvoir RDE Center ATTN: STRBE-TSE Fort Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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